BIHAR ENGINEERING UNIVERSITY, PATNA

DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING

B.Tech 4th Semester • Examination Resource

ELECTRONICS MATERIALS & INSTRUMENTATION (2024)

Exam Date: Saturday, 7 Nov 2026 (2:00 PM - 5:00 PM)

Included Question Papers

Year	Paper Title	Course Code	Examination Session
2024	Semiconductor Physics and Devices	104405	B.Tech 4th Semester Examination, 2024

Apna BEU Question Bank Archive • Compiled for B.Tech ECE 4th Semester

Contains official previous year questions with multiple-choice questions, descriptive problems, and complete circuits.

BIHAR ENGINEERING UNIVERSITY, PATNA

B.Tech 4th Semester Examination, 2024

SEMICONDUCTOR PHYSICS AND DEVICES

Time: 03 Hours | Code: 104405 | Full Marks: 70

Instructions:

1. The marks are indicated in the right-hand margin.
2. There are NINE questions in this paper.
3. Attempt FIVE questions in all.
4. Question No. 1 is compulsory.

Q.1 Choose the correct option/answer the following (Any seven question only): **[2x7=14]**

a. Which of the following best describes the energy band model of a semiconductor?

i. Electrons are localized within individual atoms
ii. Electrons occupy energy bands separated by a band gap
iii. Electrons are free to move without any restriction
iv. All energy levels are completely filled

b. In an extrinsic semiconductor at equilibrium, the Fermi level E_F shifts towards:

i. The conduction band in p-type material
ii. The conduction band in n-type material
iii. The valence band in n-type material
iv. The mid-gap energy level in doped semiconductors

c. Which of the following phenomena describe the generation and recombination processes in semiconductors?

i. Band-to-band recombination
ii. Impact ionization
iii. Auger recombination
iv. All of the above

d. What is the primary cause of the built-in potential in a PN junction diode?

i. Applied external voltage across the junction
ii. Diffusion of majority carriers
iii. Recombination of electrons and holes
iv. Ionization of donor and acceptor atoms

e. Which of the following mechanisms is responsible for junction breakdown in a Zener diode?

i. Avalanche multiplication
ii. Thermal runaway
iii. Tunneling of carriers across the junction
iv. Both (i) and (iii)

f. In which mode of operation does a Bipolar Junction Transistor (BJT) act as an amplifier?

i. Cutoff mode
ii. Active mode
iii. Saturation mode
iv. Reverse-active mode

g. What is the expression for common-base current gain (α) of a BJT?

i. $\alpha = I_C/I_B$
ii. $\alpha = I_E/I_C$
iii. $\alpha = I_C/I_E$
iv. $\alpha = I_B/I_C$

h. In a Junction Field-Effect Transistor (JFET), the drain current (I_D) in the saturation region is primarily controlled by:

i. The gate-to-source voltage (V_{GS})
ii. The source current (I_S)
iii. The drain-to-source voltage (V_{DS})
iv. The drain resistance

i. Which of the following correctly describes the threshold voltage (V_{th}) of a MOSFET?

i. The voltage at which the drain current becomes zero
ii. The voltage at which the device enters saturation
iii. The minimum gate voltage required to form an inversion layer at the semiconductor-oxide interface
iv. The voltage at which the oxide layer begins conducting current

j. Which of the following devices is primarily used for triggering other semiconductor devices like SCRs?

i. TRIAC
ii. UJT
iii. DIAC
iv. MOSFET

Q.2 Solve both questions : **[14]**

a. Explain the concept of the bonding model and energy band model in semiconductors.

b. Explain drift and diffusion current mechanisms in semiconductors. Derive the Einstein relation that connects diffusion coefficient and mobility.

Q.3 Solve both questions : **[14]**

a. Discuss the different types of recombination mechanisms and their significance in semiconductor devices.

b. Write and explain the continuity equation for charge carriers in semiconductors. Derive the minority carrier diffusion equation and discuss its importance in the operation of semiconductor devices.

Q.4 Solve both questions : **[14]**

a. Explain the Ideal Diode Equation. Derive it quantitatively using the band model and boundary conditions, stating the assumptions made in the derivation.

b. Explain the working principles and applications of special diodes such as the Zener diode, Tunnel diode, Varactor diode and Schottky diode. How do their characteristics differ from a conventional PN junction diode?

Q.5 Solve this question : **[14]**

Discuss the two primary mechanisms of junction breakdown in a PN junction diode- Avalanche breakdown and Zener breakdown. Compare their occurrence conditions and characteristics.

Q.6 Solve this question : **[14]**

Describe the impact of base width modulation (Early effect) on BJT operation. How does it affect the common-emitter current gain and output characteristics? Include relevant equations in your explanation.

Q.7 Solve this question : **[14]**

Explain the deviations from ideal MOS behaviour, including metal-semiconductor work function differences, fixed oxide charges, interface traps and threshold voltage shifts. How do these non-ideal effects influence MOSFET performance?

Q.8 Solve this question : **[14]**

Discuss the working principles and characteristics of different optoelectronic devices: PIN Photodiode, Avalanche Photodiode, Solar Cell and LED. Compare their applications.

Q.9 Solve this question : **[14]**

Explain the working of a solid-state laser diode. How does it differ from an LED in terms of construction, working principle, and applications? Discuss its advantages in optical communication.